Pratyush Padhy

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Education

University of California, Irvine — Irvine, CA

B.S. Computer Science, Specialization in Intelligent Systems, GPA: 3.8/4.0 — *Expected June 2028*

- Dean's Honor List: Fall 2025, Winter 2026, Spring 2026 | Orgs: Data@UCI, AI@UCI, Cyber@UCI, Hack@UCI

Experience

Amazon Web Services (AWS) — Palo Alto, CA

Software Development Engineer Intern — *Sept. 2026 – Present*

- Built an LLM agent that pins each Spark 4.1 regression to its exact commit by matching diffs to slowed operators
- Replaced the team's release-baseline checks with night-to-night Welch's t-tests on **1,030** runs, now its primary tool
- Shipped the agent 4 weeks early on Bedrock and ECS, cutting debugging time from **1–3 hours to 5 minutes**
- Verified **526,000+** extracted metrics per build against the team's performance reports, with zero mismatches

Tata Consultancy Services (TCS) — Pittsburgh, PA

Software Engineer Intern — *June 2026 – Aug. 2026*

- Fine-tuned the SmolVLA AI model so a simulated robot arm pours liquid using only a camera (**85.3%** success)
- Beat a Stable-Baselines3 reinforcement learning baseline (**81.5%** over 3 seeds) that saw the full simulator state
- Showed SmolVLA relies on pretraining: trained from scratch on identical input, it completed 0% of pours
- Rebuilt the simulation in Python on FluidLab in 3 weeks, fixing 3 engine bugs to unblock **1.5 months** of work

Ready Tutor — Irvine, CA

Software Developer Intern — *March 2026 – June 2026*

- Extended search-as-you-type over 1,200+ courses for Course Eater, a UCI degree planner used by **500+** students
- Cut database writes to **1 per action** by routing 10+ scattered storage calls through a React Context provider
- Shipped **30+** production fixes in TypeScript, including a credit-calculation bug that broke schedule planning

Projects

Chronicle | Python, FastAPI, PostgreSQL, Next.js | chronicles-weld.vercel.app — *June 2026 – Sept. 2026*

- Created a job aggregator ingesting **36,000+** open roles directly from 600+ Greenhouse, Lever, and Ashby boards
- Engineered hybrid full-text and pgvector search via reciprocal rank fusion, with **19ms** p95 retrieval in local tests
- Raised recommendation NDCG@10 from **0.58 to 0.81** over rule-based ranking across 24 synthetic user personas
- Improved correct pay on intern postings from **3.8% to 96.2%** by reading structured ATS pay before parsing text

Forge | Python, PyTorch, TRL, Unsloth | forge-grpo.vercel.app — *July 2026 – Sept. 2026*

- Built an end-to-end reinforcement learning pipeline that teaches Qwen2.5-1.5B-Instruct multi-step math reasoning
- Trained a LoRA adapter via GRPO, raising GSM8K pass@1 from **58.8% to 70.0%** with ARC down only 1 point
- Designed 4 verifiable rewards, fixing a zero-reward cold start with a one-shot prompt and a graded format reward
- Exported a 4-bit GGUF **3.15x** smaller than fp16 for Ollama and served base vs. tuned via vLLM multi-LoRA

FacePulse | Python, TensorFlow, OpenCV, React | face-pulse.netlify.app — *April 2026 – June 2026*

- Built a real-time computer vision app that classifies 5 emotions, **featured on the Handshake AI Showcase**
- Trained 4 TensorFlow models on FER2013, with a custom CNN (**64.5%**) matching a fine-tuned VGG16 (64.3%)
- Engineered dlib facial-landmark features that lifted an MLP from **43.6% to 51.7%** over raw pixel input

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, Scala, Bash, HTML/CSS
Machine Learning: PyTorch, TensorFlow, Hugging Face, vLLM, Ollama, Stable-Baselines3, scikit-learn, OpenCV
Data: Apache Spark, PostgreSQL (pgvector), MongoDB, SQLite, NumPy, Pandas, Matplotlib
Cloud & DevOps: AWS (S3, Lambda, ECS, Bedrock), Google Cloud, Docker, GitHub Actions, Linux, Git
Web & Testing: FastAPI, SQLAlchemy, REST APIs, React, Next.js, Tailwind CSS, pytest